Initial nurse workforce calculation

Number of nurses in Alberta: 38,900

Percentage of nurses working in hospitals: 67%

Number of nurses working in hospitals in Alberta: $38,900 \times 0.67 = 26,063$

Percentage of nurses working in COVID-unit: 26%

Initial number of nurses working in COVID-unit in Alberta: $26,063 \times 0.26 = 6776$

Initial number of nurses working in other-unit in Alberta: $26,063 - 6776 = 19,287$

Table 1: Weekly community infection risk calculation

	Alpha	**Omicron**
Community seroprevalence from data	August 2020: 0.00749 December 2020: 0.01996 Difference: 0.01247	December 2021: 0.1294 February 2022: 0.2489 Difference: 0.1195
Weekly increase in community seroprevalence	$\int_{20}^{40} mx\,dx = 0.01247$ $m = 2.0783 \times 10^{-5}$	$\int_{91}^{97} mx\,dx = 0.1195$ $m = 2.1188 \times 10^{-4}$
Weekly community infection risk	$\theta_{20} = \int_{20}^{21} mx\,dx = 0.0004260515$ $\theta_{21} = \int_{21}^{22} mx\,dx = 0.0004468345$ $\theta_{22} = \int_{22}^{23} mx\,dx = 0.0004676175$ $\theta_{23} = \int_{23}^{24} mx\,dx = 0.0004884005$ $\theta_{24} = \int_{24}^{25} mx\,dx = 0.0005091835$ $\theta_{25} = \int_{25}^{26} mx\,dx = 0.0005299665$ $\theta_{26} = \int_{26}^{27} mx\,dx = 0.0005507495$ $\theta_{27} = \int_{27}^{28} mx\,dx = 0.0005715325$ $\theta_{28} = \int_{28}^{29} mx\,dx = 0.0005923155$ $\theta_{29} = \int_{29}^{30} mx\,dx = 0.0006130985$ $\theta_{30} = \int_{30}^{31} mx\,dx = 0.0006338815$ $\theta_{31} = \int_{31}^{32} mx\,dx = 0.0006546645$	$\theta_{91} = \int_{91}^{92} mx\,dx = 0.01938822$ $\theta_{92} = \int_{92}^{93} mx\,dx = 0.01959590$ $\theta_{93} = \int_{93}^{94} mx\,dx = 0.01980338$ $\theta_{94} = \int_{94}^{95} mx\,dx = 0.02001006$ $\theta_{95} = \int_{95}^{96} mx\,dx = 0.02021674$ $\theta_{96} = \int_{96}^{97} mx\,dx = 0.02042342$

	$\theta_{32} = \int_{32}^{33} mx\,dx = 0.0006754475$ $\theta_{33} = \int_{33}^{34} mx\,dx = 0.0006962305$ $\theta_{34} = \int_{34}^{35} mx\,dx = 0.0007170135$ $\theta_{35} = \int_{35}^{36} mx\,dx = 0.0007377965$ $\theta_{36} = \int_{36}^{37} mx\,dx = 0.0007585795$ $\theta_{37} = \int_{37}^{38} mx\,dx = 0.0007793625$ $\theta_{38} = \int_{38}^{39} mx\,dx = 0.0008001455$ $\theta_{39} = \int_{39}^{40} mx\,dx = 0.0008209285$	

Table 2: Weekly Costs for Alpha (Aug - Dec 2020)

Week	Total Infected Nurses	Total Cost for 0% Redeployment	Total Cost for 50% Redeployment	Total Cost for 100% Redeployment
20-21	2.8869	8206.879	6839.066	5471.253
21-22	4.7925	13624.12	11353.43	9082.746
22-23	6.0957	17328.86	14440.71	11552.57
23-24	7.0312	19988.3	16656.91	13325.53
24-25	7.7419	22008.67	18340.56	14672.45
25-26	8.3131	23632.48	19693.73	15754.99
26-27	8.8007	25018.63	20848.86	16679.09
27-28	9.2426	26274.86	21895.72	17516.58
28-29	9.6556	27448.94	22874.12	18299.29
29-30	10.0574	28591.18	23825.98	19060.78
30-31	10.4596	29734.55	24778.79	19823.03
31-32	10.8546	30857.46	25714.55	20571.64
32-33	11.2523	31988.04	26656.7	21325.36

33-34	11.6613	33150.74	27625.62	22100.5
34-35	12.0990	34395.04	28662.53	22930.02
35-36	12.5667	35724.61	29770.51	23816.41
36-37	13.0801	37184.11	30986.76	24789.41
37-38	13.6632	38841.74	32368.12	25894.5
38-39	14.3019	40657.44	33881.2	27104.96
39-40	14.9551	42514.36	35428.63	28342.91
Total Cost		567171	472642.5	378114

Table 3: Weekly Costs for Omicron (Dec 2021 – Feb 2022)

Week	Total Infected Nurses	Total Cost for 0% Redeployment	Total Cost for 50% Redeployment	Total Cost for 100% Redeployment
91-92	131.3746	373471.7	311226.4	248981.1
92-93	214.2678	609120.5	507600.4	406080.3
93-94	267.4039	760175.8	633479.8	506783.9
94-95	303.8044	863655.1	719712.6	575770.1
95-96	332.3866	944908.6	787423.9	629939.1
96-97	357.2265	1015523	846269.6	677015.7
Total Cost		4566855	3805713	3044570

Table 4: Weekly Infected and Backfill Nurses (0% Redeployment, Alpha):

Week	Total Infected Nurses	Total Overtime Hours	Nurses Required for 4 hours Overtime	Nurses Required for 8 hours Overtime	Nurses Required for 20 hours Overtime
20-21	2.89	115	29	14	6
21-22	5.91	192	48	24	10
22-23	6.2	244	61	30	12
23-24	6.48	281	70	35	14
24-25	6.76	310	77	39	15
25-26	7.04	333	83	42	17

26-27	7.32	352	88	44	18
27-28	7.6	370	92	46	18
28-29	7.89	386	97	48	19
29-30	8.17	402	101	50	20
30-31	8.45	418	105	52	21
31-32	8.73	434	109	54	22
32-33	9.01	450	113	56	23
33-34	9.29	466	117	58	23
34-35	9.58	484	121	60	24
35-36	9.86	503	126	63	25
36-37	10.14	523	131	65	26
37-38	10.42	547	137	68	27
38-39	10.7	572	143	72	29
39-40	10.98	598	150	75	30

Table 5: Weekly Infected and Backfill Nurses (0% Redeployment, Omicron):

Week	Total Infected Nurses	Total Overtime Hours	Nurses Required for 4 hours Overtime	Nurses Required for 8 hours Overtime	Nurses Required for 20 hours Overtime
91-92	131.37	5255	1314	657	263
92-93	264.17	8571	2143	1071	429
93-94	267.04	10696	2674	1337	535
94-95	269.91	12152	3038	1519	608
95-96	272.78	13295	3324	1662	665
96-97	275.65	14289	3572	1786	714

Table 6: Weekly Infected and Backfill Nurses (50% Redeployment, Alpha):

Week	Total Infected Nurses	Total Overtime Hours	Nurses Required for 4 hours Overtime	Nurses Required for 8 hours Overtime	Nurses Required for 20 hours Overtime
20-21	2.89	58	14	7	3
21-22	5.91	96	24	12	5
22-23	6.2	122	30	15	6
23-24	6.48	141	35	18	7
24-25	6.76	155	39	19	8
25-26	7.04	166	42	21	8

26-27	7.32	176	44	22	9
27-28	7.6	185	46	23	9
28-29	7.89	193	48	24	10
29-30	8.17	201	50	25	10
30-31	8.45	209	52	26	10
31-32	8.73	217	54	27	11
32-33	9.01	225	56	28	11
33-34	9.29	233	58	29	12
34-35	9.58	242	60	30	12
35-36	9.86	251	63	31	13
36-37	10.14	262	65	33	13
37-38	10.42	273	68	34	14
38-39	10.7	286	72	36	14
39-40	10.98	299	75	37	15

Table 7: Weekly Infected and Backfill Nurses (50% Redeployment, Omicron):

Week	Total Infected Nurses	Total Overtime Hours	Nurses Required for 4 hours Overtime	Nurses Required for 8 hours Overtime	Nurses Required for 20 hours Overtime
91-92	131.37	2627	657	328	131
92-93	264.17	4285	1071	536	214
93-94	267.04	5348	1337	669	267
94-95	269.91	6076	1519	760	304
95-96	272.78	6648	1662	831	332
96-97	275.65	7145	1786	893	357

Cost Analysis

During Alpha, the number of infected nurses increased gradually from 2.9 to 15 FTEs. Weekly costs rose accordingly, with total expenditures over the 20-week period amounting to:

- 0% redeployment (all overtime): 567,171 CAD
- 50% redeployment: 378,114 CAD and
- 100% redeployment: 330,850 CAD

The results demonstrate that redeployment substantially reduces costs compared to reliance on overtime alone.

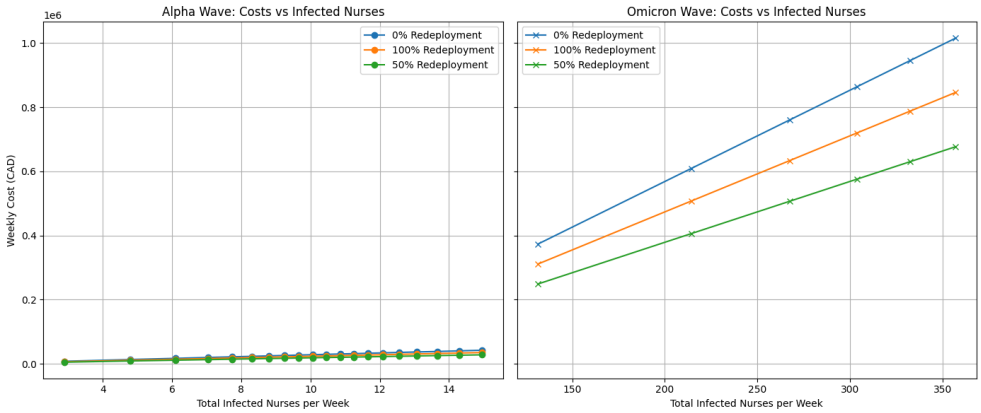
During Omicron, the number of infected nurses increased sharply, rising from 131 to 357 full-time equivalents (FTEs). This escalation led to a substantial rise in weekly staffing costs, with total expenditures reaching

- 0% redeployment (all overtime): 4,566,855 CAD
- 50% redeployment: 3,044,570 CAD and
- 100% redeployment: 2,663,999 CAD

These results demonstrate the significant financial strain hospitals face during severe epidemic surges. While redeployment strategies consistently reduced costs compared to reliance on overtime alone, they also required shifting staff away from non-COVID units, which may limit capacity in other areas of hospital operations.

The detailed weekly cost tables for both waves which show infected nurse counts and total weekly expenditures under each staffing scenario, are provided in the Appendix (Table 2,3) for reference. These tables enable direct comparison across the 0%, 50%, and 100% redeployment strategies and illustrate how costs evolve as weekly infections increase.

The following two figures illustrate the weekly staffing costs for Wave 1 and Wave 2 under different redeployment strategies. Costs increase with the number of infected nurses, with overtime-only coverage resulting in the highest expenses, full redeployment the lowest, and a 50% mixed strategy providing an intermediate, more balanced cost outcome.



<u>Weekly Overtime Burden: Number of Nurses Needed</u>

Overtime limits strongly affect how many nurses are needed to cover absences. With a 4-hour cap, 10 nurses are required to replace one full-time equivalent (FTE), with an 8-hour cap, 5 nurses are needed and with a 20-hour cap, only 2 nurses are sufficient.

For 0% Redeployment and 100 % Overtime:

Total Overtime Hours	$I_t \times 40$ hours
Total nurses required for 4 hours overtime	$I_t \times 10$ nurses
Total nurses required for 8 hours overtime	$I_t \times 5$ nurses
Total nurses required for 20 hours overtime	$I_t \times 2$ nurses

During Alpha Wave, the number of infected nurses increased gradually from fewer than 3 to about 11 FTEs. Depending on the overtime cap, between 29 and 150 nurses were required to share the load at 4 hours/week, compared with 6 to 30 nurses at 20 hours/week. In the Omicron Wave, infection levels rose much more sharply, from 131 to 276 FTEs. Overtime requirements escalated to over 14,000 hours per week. At the strictest cap (4 hours/week), more than 3,500 nurses were needed to absorb the overtime, while at 20 hours/week the requirement dropped to fewer than 750 nurses.

For 50% Redeployment and 50 % Overtime:

In this scenario, half of the required nurses will be backfilled from the nurses as FTE from another unit.

Total Overtime Hours	$I_t \times 20$ hours
Total nurses required for 4 hours overtime	$I_t \times 5$ nurses
Total nurses required for 8 hours overtime	$I_t \times 2.5$ nurses
Total nurses required for 20 hours overtime	$I_t \times 1$ nurse

A 50% redeployment scenario reduced the overtime burden by half, since part of the backfill was covered by full-time redeployed nurses. Even so, the scaling effect remained as lower overtime caps required substantially more nurses to share the workload, while higher caps concentrated the burden on fewer individuals.

Detailed weekly values for both waves and scenarios are provided in the Appendix (Tables 4-7). These results underscore that while redeployment reduces financial costs and overtime hours, the choice of overtime cap strongly affects workforce distribution and the risk of burnout.